

Format description for the IVS Pilot Project - Troposphere Product

-) Header and Footer Lines (Mandatory)

Description

The Header line must be the first line in a SINEX_TRO file.
The Footer line must be the last line in a SINEX_TRO file.

Contents:

H E A D E R L I N E		
Field	Description	Format
File Identifier	%=TRO	A5
Format Version	Four digits indicating the version of SINEX_TROP format used. IVS PP: 9.99	1X,F4.2
File Agency Code	Identify the agency creating the file.	1X,A3
Time	Creation time of this SINEX file.	1X,I2.2, ':', I3.3, ':', I5.5
Agency Code	Identify the agency providing the data in the SINEX file	1X,A3
Time	Start time of solution in the this SINEX_TROP file	1X,I2.2, ':', I3.3, ':', I5.5
Time	End time of the solution in the this SINEX_TROP file	1X,I2.2, ':', I3.3, ':', I5.5
Observation Code	Technique(s) used to generate the SINEX solution IVS PP: 'R'	1X,A1
Solution Contents	Marker name if this is a combined solution file and contains only one site or 'MIX' if it is a submission file containing more than one site IVS PP: 'MIX'	1X,A4
		64

F_O_O_T_E_R_L_I_N_E		
Field	Description	Format
File Identifier	%=ENDTRO	A8
		8

-) FILE/REFERENCE Block (Mandatory)

Description:

This block provides information on the Organization, point of contact, the software and hardware involved in the creation of the file.

Contents:

F_I_L_E_R_E_F_E_R_E_N_C_E_D_A_T_A_L_I_N_E		
Field	Description	Format
Information Type	Describes the type of information present in the next field. May take on the following values: 'DESCRIPTION' - Organization(s) gathering/altering the file contents. 'OUTPUT' - Description of the file contents. 'CONTACT' - Address of the relevant contact. e-mail 'SOFTWARE' - Software used to generate the file. 'HARDWARE' - Computer hardware on which above software was run. 'INPUT' - Brief description of the input used to generate this solution. Any of the above fields may be and in any order.	1X,A18
Information	Relevant information for the type indicated by the previous field.	1X,A60
		80

-) SITE/ID Block

Description:

This block provides general information for each site containing estimated parameters.

For DOMES numbers and station description please refer to
ftp://lareg.ensg.ign.fr/pub/itrf/iers_dir.sta

If you use the first four letters of the 8-letter station name
(<ftp://ivscc.gsfc.nasa.gov/pub/control/ns-codes.txt>)
this block can be skipped.

Use the minus sign for negative approximate longitude or latitude only in the 'degrees' component and do not repeat it in the 'minutes' and 'seconds' component.

Contents:

S I T E I D D A T A L I N E		
Field	Description	Format
[Site Code]	Call sign for a site. IVS PP: first four letters of the 8-letter station name	1X,A4
[Point Code]	Physical monument used at a site	1X,A2
Unique Monument Identification	Unique alpha-numeric monument identification. For ITRF purposes, it is a nine character DOMES/DOMEX number (five/six digits, followed by the single letter 'M' or 'S', followed by four/three digits)	1X,A9
[Observation Code]	Observation technique(s) used. IVS PP: 'R'	1X,A1
Station Description	Free-format description of the site, typically the town and/or country.	1X,A22
Approximate Longitude	Approximate longitude of the site in degrees(W/+), minutes and seconds.	1X,I3, 1X,I2, 1X,F4.1
Approximate Latitude	Approximate latitude of the site in degrees(NS/+-), minutes and seconds.	1X,I3, 1X,I2, 1X,F4.1
Approximate	Approximate height of the site in	1X,F7.1

Height	metres.	
		75

-) SITE/ECCENTRICITY Block

Description:

List of antenna eccentricities from the Marker to the Antenna Reference Point (ARP) or to the intersection of axis.

If you use the eccentricities as described in http://gemini.gsfc.nasa.gov/solve_save/ECCDAT.ecc this block can be skipped.

Contents:

S I T E E C C E N T R I C I T Y D A T A L I N E		
Field	Description	Format
[Site Code]	Site code for which some parameters are estimated.	1X,A4
[Point Code]	Point Code at a site for which some parameters are estimated.	1X,A2
[Solution ID]	Solution ID at a Site/Point code for which some parameters are estimated.	1X,A4
[Observation Code]	Identification of the observation technique used.	1X,A1
[Time]	Time since the antenna has been installed at the Site/Point. Value 00:000:00000 indicates that the antenna has been installed at least since the "File Epoch Start Time".	1X,I2.2, 1H:,I3.3, 1H:,I5.5
[Time]	Time until the antenna is installed at a Site/Point. Value 00:000:00000 indicates that the antenna has been installed at least until the "File Epoch End Time".	1X,I2.2, 1H:,I3.3, 1H:,I5.5
Eccentricity Reference System	Reference system used to describe vector distance from monument benchmark to the antenna reference point: 'UNE' - Local reference system	1X,A3

	Up, North, East. 'XYZ' - Cartesian Reference System X, Y, Z. All units are in meters.	
Up / X Eccentricity	Up / X offset from the marker to the Antenna reference point (ARP).	1X,F8.4
North / Y Eccentricity	North/Y offset from the marker to the Antenna reference point (ARP).	1X,F8.4
East / Z Eccentricity	East/Z offset from the marker to the Antenna reference point (ARP).	1X,F8.4
		72

-) TROP/STA_COORDINATES Block (Mandatory)

Description

This block provides the coordinates of the sites. For the combination result it also gives some statistical information.

Contents:

TROP/STA_COORDINATES_D_A_T_A_L_I_N_E		
Field	Description	Format
Site Code	Call sign for a site. IVS PP: Site Code as discribed in Site/ID or first four letter of 8-letter station name	1X,A4
Point Code	Physical monument used at a site	1X,A2
Solution ID	Solution number to which the input in this data line is referred to.	1X,A4
Observation Code	Observation technique used.	1X,A1
Coordinates	x,y,z-coordinate of a site (a priori plus estimated)	3(1X,F12.3)
System	Terrestrial Reference System Code	1X,A6
Remark	A remark used to identify the origin of the coordinates	1X,A5

	(AC acronym or 'Mean')	
Standard Deviation	Standard deviation for x,y,z in [mm] (Used only for Mean)	3(1X,I2)
Counter	Number of ACs used for Mean (Used only for Mean)	1X,I2
		79

-) TROP/DESCRIPTION Block (Mandatory)

Description

This block gives important parameters from the analysis and defines the fields in the block 'TROP/SOLUTION'.

Contents:

TROP/DESCRIPTION__D_A_T_A__L_I_N_E		
Field	Description	Format
Information Type	Describes the type of information present in the next field. May take one the following values: 'SOLUTION_FIELDS_1': (7(1X,A6)) Names of fields in solution (see below) 'SOLUTION_FIELDS_2': (7(1X,A6)) Used if more than 7 fields must be coded 'ELEVATION CUTOFF ANGLE': (1X,F22) elevation cutoff angle in [deg] 'DRY MAPPING FUNCTION' (1X,A22) 'WET MAPPING FUNCTION' (1X,A22) 'DOWNWEIGHTING OF LOW OBS.': YES(/NO) if observations at low elevations are (/not) downweighted (1X,A22) if YES then ... 'DOWNWEIGHTING' (1X,A22) 'TERRESTRIAL REFERENCE FRAME' (1X,A22) 'TRF FIXED': (1X,A22) YES/ NO/ OTHER 'SAMPLING ZENITH DELAYS': (1X,I22) sampling time interval for zenith delays in [s] 'DRY DELAY MODEL': (1X,A22) model to calculate dry delay 'WET DELAY MODEL': (1X,A22) model to calculate apriori wet delay	1X,A29

	'ZENITH DELAY MODEL': (1X,A22) functional model 'CONSTR. ZENITH DELAY RATE' (1X,A22) 'CONSTR. ZENITH DELAY OFFSET' (1X,A22) 'SAMPLING GRADIENTS': (1X,I22) sampling time interval for gradients in [s] 'APRIORI NORTH GRADIENT': (1X,F22) apriori value for north gradient in [mm] 'APRIORI EAST GRADIENT': (1X,F22) apriori value for east gradient in [mm] 'GRADIENT MODEL': (1X,A22) functional model 'CONSTR. GRADIENT RATE' (1X,A22) 'CONSTR. GRADIENT OFFSET' (1X,A22) The above fields may be in any order.	
Information	Relevant information for the type indicated by the previous field.	format is type-dependent
		53

Fields TROTOT - Total zenith path delay [mm]
 TROWET - Wet zenith path delay [mm]
 GRTOTN - Total north gradient in [mm]
 GRTOTE - Total east gradient in [mm]
 GRHYDN - (a priori) north hydrostatic gradient in [mm]
 GRHYDE - (a priori) east hydrostatic gradient in [mm]
 GRWETN - Wet north gradient in [mm]
 GRWETE - Wet east gradient in [mm]
 STDDEV - stddev of the values in the column before

-) TROP/SOLUTION Block (Mandatory)

Description

This block contains the solution for all epochs.
 Contents:

TROP/SOLUTION__D_A_T_A__L_I_N_E		
Field	Description	Format
Marker	Name of the marker	1X,A4
Time	Time epoch of the solution (*)	1X,I2.2, ':', I3.3, ':', I5.5
Values	Space separated fields of variable	no format

	length . Number and order of fields are given in the block TROP/DESCRIPTION Readable by: read(line(20:),*)(val(i),i=1,n)	
		variable <80

(*) The data intervals should be 60 minutes
starting at 18:00 UT (smaller intervals, which are
integer parts of 60 minutes are possible)